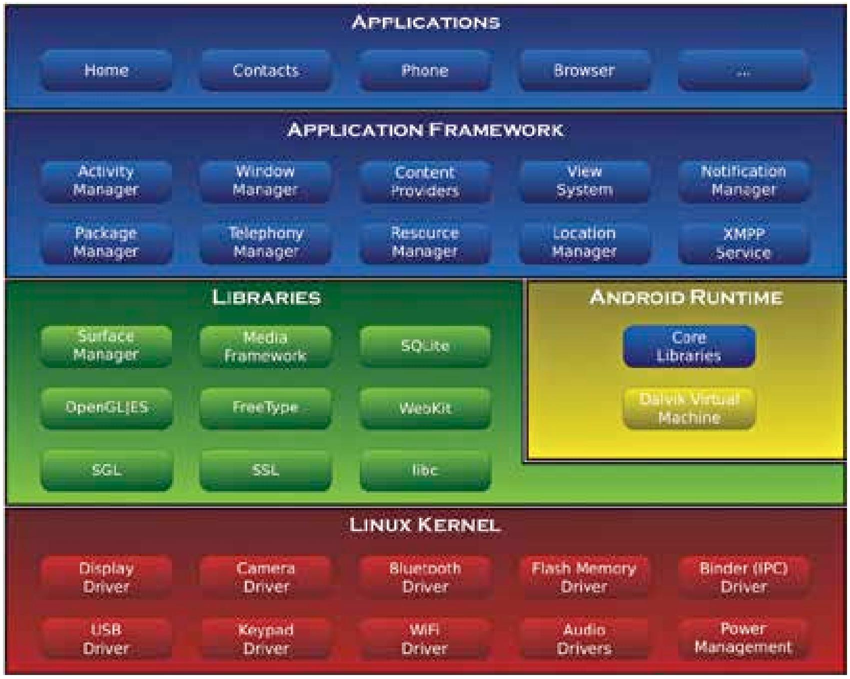


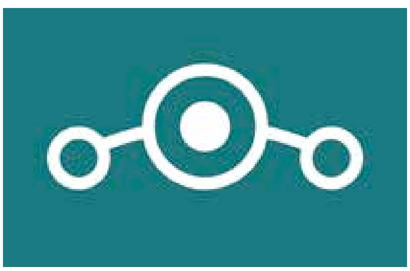
The structure of the Android Open Source Project

ibility that you see so many different implementations and customisations for Android – from smartphone manufacturers to hobbyists. A particular customised implementation is generally referred to as a Custom ROM.

So can you get into customising Android? Why not? We're here to show you how. But before we begin, there are a few things you need to know.

DISCLAIMER

Customising Android is not entirely simple. But that's not unexpected, considering all the wonderful things you can do with your Android phone now. So, as a consequence, there are a few things that you need to know to be able to do this successfully. Broadly speaking, you need to have some basic coding skills, some experience with the Linux based command line and the concept of a makefile.



While we are customising, it will be a while before you can build the next Cyanogen or LineageOS

A makefile is a document that contains the directives that work when the make command is issued. These are essentially instructions about how to compile a particular file. For instance, when a file has to be recompiled on the basis of particular changes being saved, that is an instruction you will find in the makefile.

Another disclaimer: we are not going to build the next Cyanogen Mod here. As we said earlier, it is not easy to build an entire operating system and the output of this section is going to be minor changes at best. The rest we leave up to you to learn. You will need a boatload of patience, but it will be rewarding in the end.

REQUIREMENTS

Before we start, there are a few more things that need to be decided. First of all, Android can be theoretically built for any modern-day computing system. That being said, customisation references and resources are more likely to be available more easily for devices that support customisation out-of-the-box. The demo build that we used as a reference for this article uses a Nexus 5X for the customisation. However, you can do a lookup if there are any particular restrictions or requirements for the specific model of your phone.

You'll also need access to a Linux machine or a Mac. In terms of specifications, you'll need about 130-150GB of storage and quite a lot of RAM (upto 16GB on Linux) – we never said Android was light. And regarding the overall time it will take, don't expect it to be done in minutes.

OVERVIEW OF BUILD PROCESS

There are two main steps to this process – downloading the Android SDK and then modifying the source code implementation as a custom ROM.

It is most certainly not possible to complete the entire process in this one article. Something like that is more suited to an entire *Fast Track*. We will be highlighting the broad steps involved though the best place to start off would be Google's documentation on the Android Open Source Project. You can find that at http://dgit.in/AOSPDoc. Take the documentation seriously as each point in there is important for a successful build.

Moving on, here are the broad tasks that you'll need to do, along with some pro tips for each:

- Set up a build environment – including installing the right development tools, the Java Development Kit, and getting all the paths and directories right.
  - **Pro Tip:** Ubuntu 14.04 is the recommended build environment for Linux users and OS X 10.11 for Mac users. You need to install OpenJDK 8 on Linux and Oracle's JDK 8 on OS X. On OS X you also need Macports installed along with Xcode an the Xcode command line tools.

For the keyboard warriors

Stock keyboards are a bore, so we've got some amazing replacement keyboards for your phone. Just follow this link:

http://dgit.in/NinjaTxt